



Practical - 26

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Code:

```
import breeze.linalg._
import breeze.numbers._

object Scala25 extends App {
  // Dataset
  val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0)

  // Binary labels
  val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0)

  // Initial weight and bias
  var weight = 0.0
  var bias = 0.0

  // Sigmoid function
  def sigmoid(z: Double): Double = {
    1.0 / (1.0 + math.exp(-z))
  }

  // Train the model
  for (i <- 1 to 1000) {
    for (j <- 0 until x.length) {
```

```

    val prediction = sigmoid(weight * x(j) + bias)
    val error = prediction - y(j)

    weight = weight - 0.01 * error * x(j)
    bias = bias - 0.01 * error
  }
}

// Classify the data
println("Predictions:")

for (i <- 0 until x.length) {
  val probability = sigmoid(weight * x(i) + bias)
  val result = if (probability >= 0.5) 1 else 0

  println(
    "Input = " + x(i) +
    ", Probability = " + probability +
    ", Class = " + result
  )
}

```

Output:

```

Predictions:
Input = 1.0, Probability = 0.0515501057576029, Class = 0
Input = 2.0, Probability = 0.16412826954644827, Class = 0
Input = 3.0, Probability = 0.4149888799701166, Class = 0
Input = 4.0, Probability = 0.7193154698480789, Class = 1
Input = 5.0, Probability = 0.9025176083649726, Class = 1

Process finished with exit code 0

```